

Cycle vs event rebuild time for three planners

A typical rebuild is far below the 100 ms limit (95 % of rebuilds finish within 10–30 ms). No rebuild exceeded the 100 ms limit (slowest 63 ms), so no robot was stopped; event triggering still needs 5–11× fewer rebuilds than cycling at 20 ms checks. Rebuild time is mostly map construction; the Dijkstra search is a few percent. Times are wall clock on one machine, successful rebuilds only. ▲ blue (bold) = better of the pair, ▼ orange = worse; times marked only when they differ by 10 % or more.

	Rebuilds per run	Time per rebuild, median (ms)	95 % of rebuilds under (ms)	99 % of rebuilds under (ms)	Slowest rebuild (ms)	Rebuilds over 100 ms per 10,000	Share of time in Dijkstra search (%)	Robots stopped by 100 ms limit
Voronoi								
Cycle @ 20 ms	81.2 ▼	8.7	10.3	12.0	22	0	1	0
Event (ours) @ 20 ms	7.1 ▲	8.7	10.3	11.4	20	0	1	0
Cycle @ 100 ms	15.4 ▼	8.8	10.3	11.8	24 ▼	0	1	0
Event (ours) @ 100 ms	4.1 ▲	8.9	10.5	11.9	15 ▲	0	1	0
PRM								
Cycle @ 20 ms	78.6 ▼	23.3	29.8	32.9	63 ▼	0	1	0
Event (ours) @ 20 ms	7.7 ▲	23.5	30.0	34.9	54 ▲	0	1	0
Cycle @ 100 ms	15.2 ▼	23.4	30.1	33.2	54	0	1	0
Event (ours) @ 100 ms	4.6 ▲	23.2	30.1	33.5	56	0	1	0
Visibility Graph								
Cycle @ 20 ms	119.7 ▼	20.4	25.7	28.9	54 ▼	0	3	0
Event (ours) @ 20 ms	26.5 ▲	21.0	26.0	29.0	46 ▲	0	3	0
Cycle @ 100 ms	21.4 ▼	20.1	25.3	27.8	42 ▼	0	3	0
Event (ours) @ 100 ms	8.4 ▲	20.6	26.0	28.3	34 ▲	0	3	0